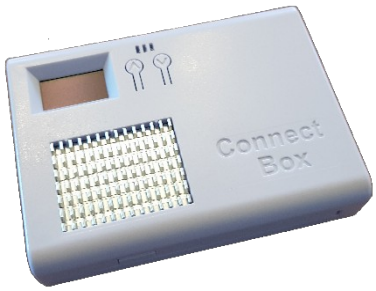
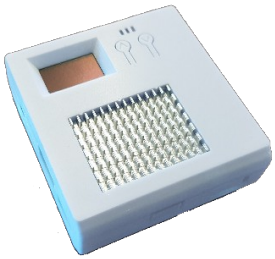
Connectbox Quick Start Guide

Up to Firmware Version 20241008



Model P V6
P/N 10001

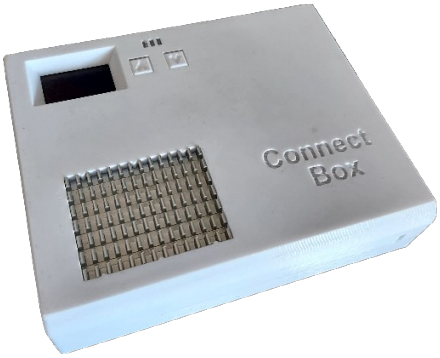


Model S V6
P/N 10002

Model P2 V7
P/N 10011

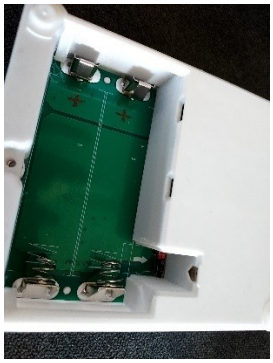
Model S2 V7
P/N 10012

Model P3 V8
P/N 10020:





Removable 21700 Batteries



Removable uSD under small cover



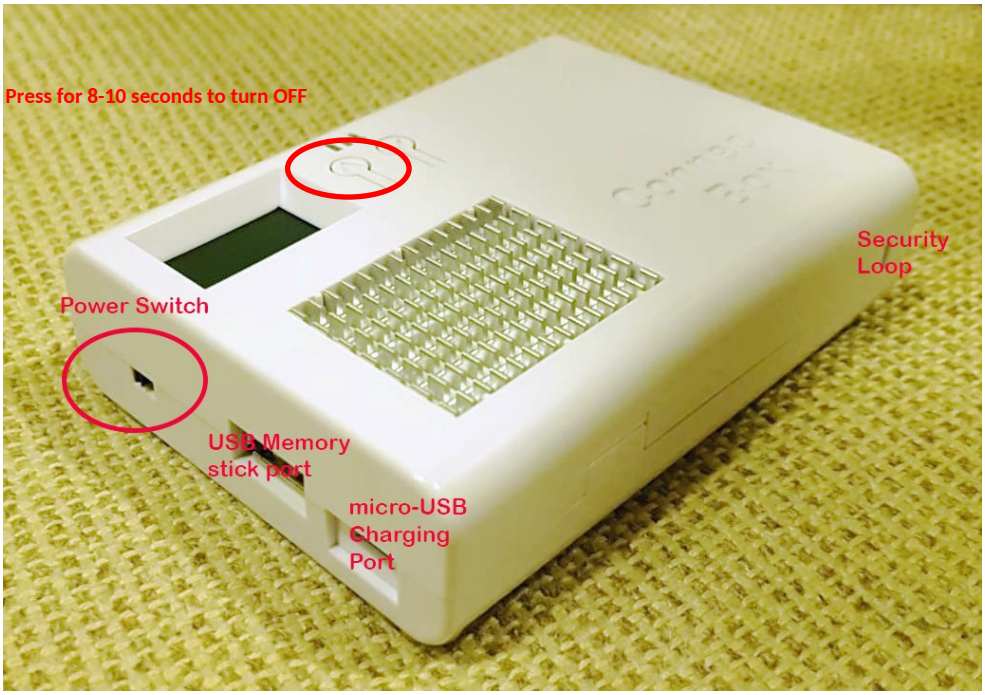
18650's with spacing adapters

Step 1: Charging your Connectbox

When you first receive the connectbox it must be charged to full capacity before using it. For a model R removable battery unit first insert two 21700 batteries taking care to see that + and – are correctly positioned. If your using the model R removable battery unit with 18650's place the Red adapter cap over the + side of the battery and the Green adapter cap over the – side and insert them into the connectbox taking care to see that t + and – are correctly positioned.

Then plug in the power module that came with the unit and using the supplied cable connect the power module to the connectbox. You should see the RED led illuminate indicating that it is charging. When the batteries are fully charged the RED LED will not be illuminated. If the RED led flashes, it indicates that the batteries are below the minimum suggested voltage level, and they will be trickle charged until they reach 3.2V respectively. After this happens the RED led should show a solid red color until they are fully charged. If after 12 hours the flashing RED led continues, the batteries in the unit are not usable and should be changed.

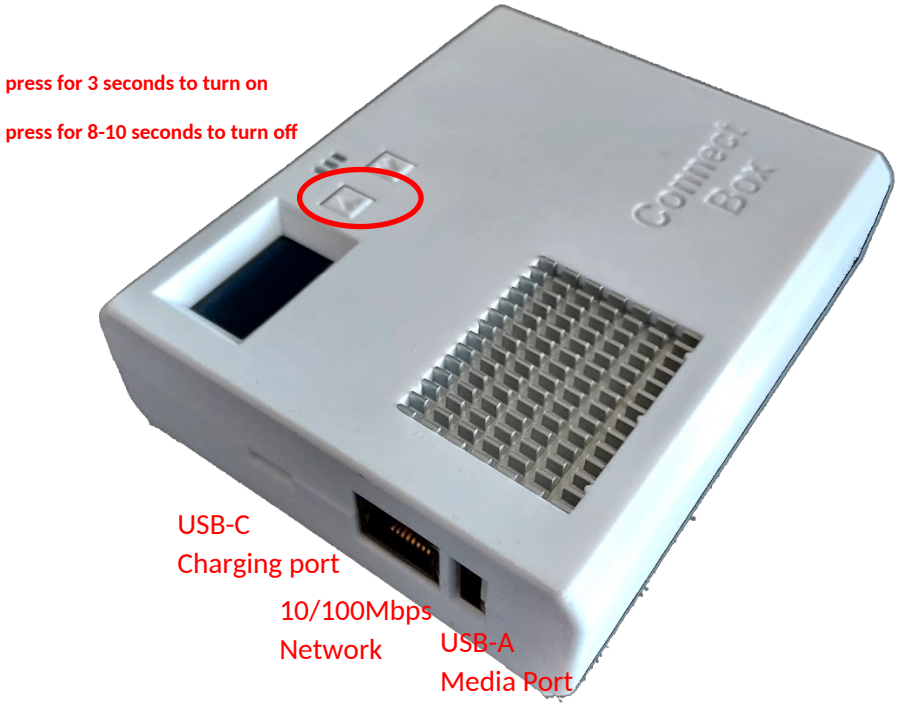
Step 2: Turning the Connectbox on or off.



Connectbox Model V6 Hardware



Connectbox Model V7



Connectbox Model V8 Hardware

With the USB-C charger plugged in or with a relatively full battery, turn your unit on using the up-arrow switch on the top of the unit top. Allow a **minute or two** to pass before using a wireless device to locate and join the default wireless network with the default called *ConnectBox - Free Media*. You will know when it is available by the **orange** LED turning on. (Note: that if you charge your unit and run the unit simultaneously in a hot climate it may cause a red flashing indicator of high temperature internal to the unit.

To turn on the Model V6 switch the side slide switch on or off-on. For the Model V8 hardware press the up arrow for 3 to 5 seconds to turn on. For both models V6 and V8 the up arrow can turn the unit off if pressed and held for 8 or more seconds. This is the preferred way to turn the unit off. You will know if the unit is on by the **Green** LED

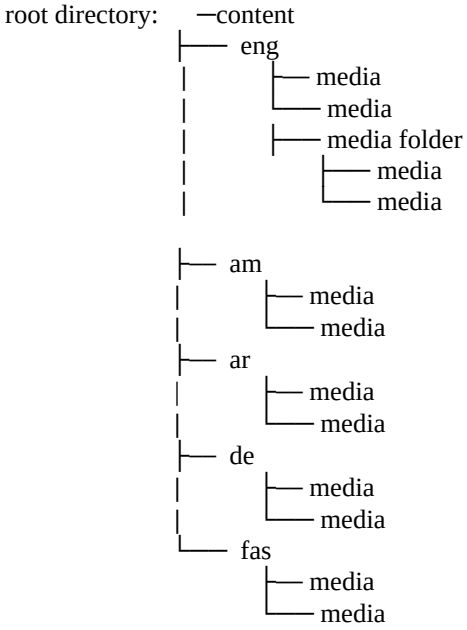
turning on. Additionally, the up arrow and down arrow can be used after the unit is turned on to cycle through the screen menus.

Step 3: Loading media onto your connectbox.

You have the option of connecting an external USB flash drive to the USB A port that is available on the Connectbox' s exterior. The USB A port will in a sense override any content that is stored on the internal microSD card within the unit. The USB stick should be formatted for EXT4, FAT32 or NTFS format. We recommend a slim line USB media stick like the SanDisk Ultra Fit line for low profile and USB 3 speed is recommended for faster system operation. Disc capacities from 4 GB up to 512GB can be used. Alternatively, you can copy the contents of a USB A flash drive and thus store it on the internal uSD card. Generally, there is about 54GB of space available for content on the internal uSD card.

Preparing content for the Connectbox on a USB A flash drive, when media is stored on the connectbox external USB flash drive or internal uSD flash drive, the Connectbox will create a user interface to the content. The root level of the storage is used to define if this is a single language media set or a multi-language media set. To use multiple languages on the Connectbox, the root level of the media storage must have folders titled with the ISO 639-2 code of the language's. All the specific language media is then held under that folder:

This example would hae a folders labeled:



Folders at the content directory that do not contain valid ISO language codes will be ignored. There also should be no media at the content level if language folders are used as it will be ignored. Note that either 2 character or 3 character ISO codes can be used. The above would be prepared to have 5 languages, English, Amharic, Arabic, German and Persian.

If you only want one languge then the content can be loaded directly into the root directory under the content folder. Additionally, you can add a file named **.language** with the contents containing the ISO code of the language you are using. Note the period befor the name is important.

```
Root directory    - content
                  ├── media
                  ├── media
                  ├── media
                  ├── media folder
                  │       ├── media
                  │       └── media
                  └── .language (with content  containing a single ISO code)
```

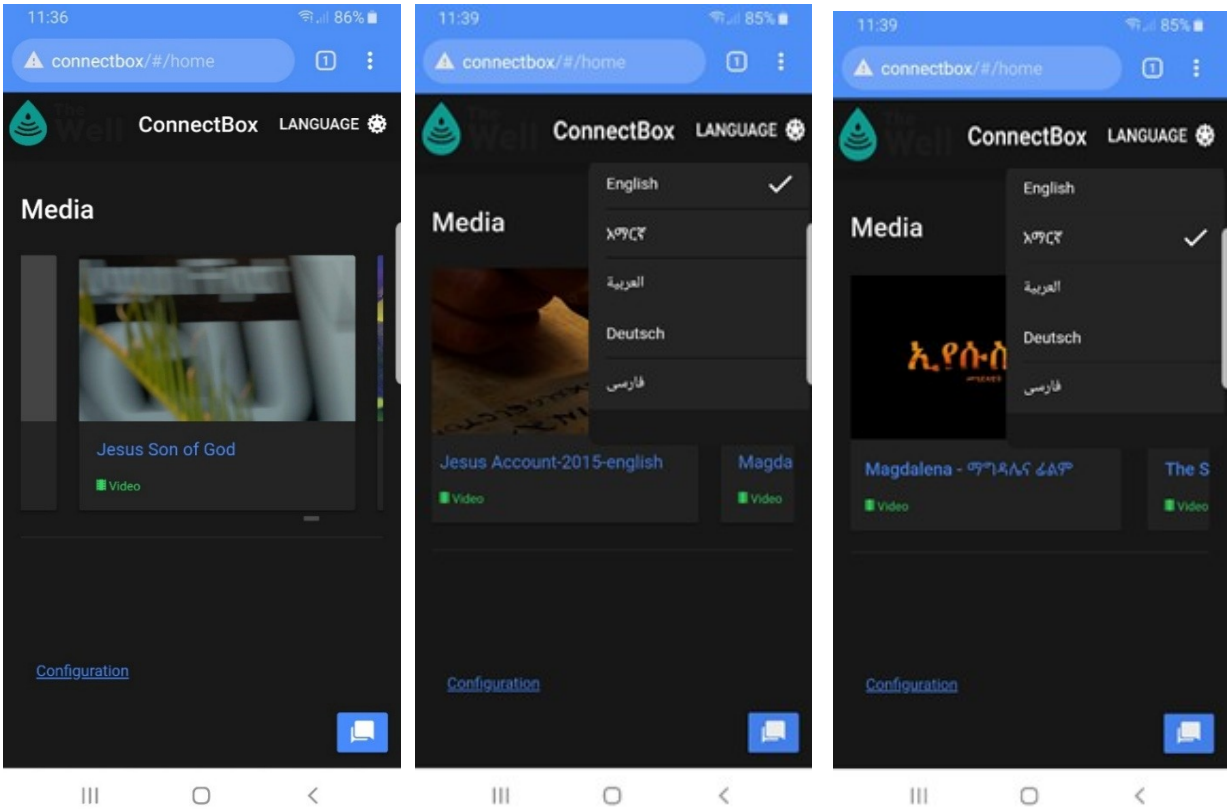
The media content can be any of the following types: mp4, pdf, png, jpeg, mp3, apng, avif, gif, svg+xml, webp, webm, ogg, etc. See the detailed support documentation on the browsers to get an understanding of the extensions of the usable media and codecs used in different media type.

Once the media is inserted the Connectbox when powered up will create an interface to the media. This includes generic icons for media files as well as collections:

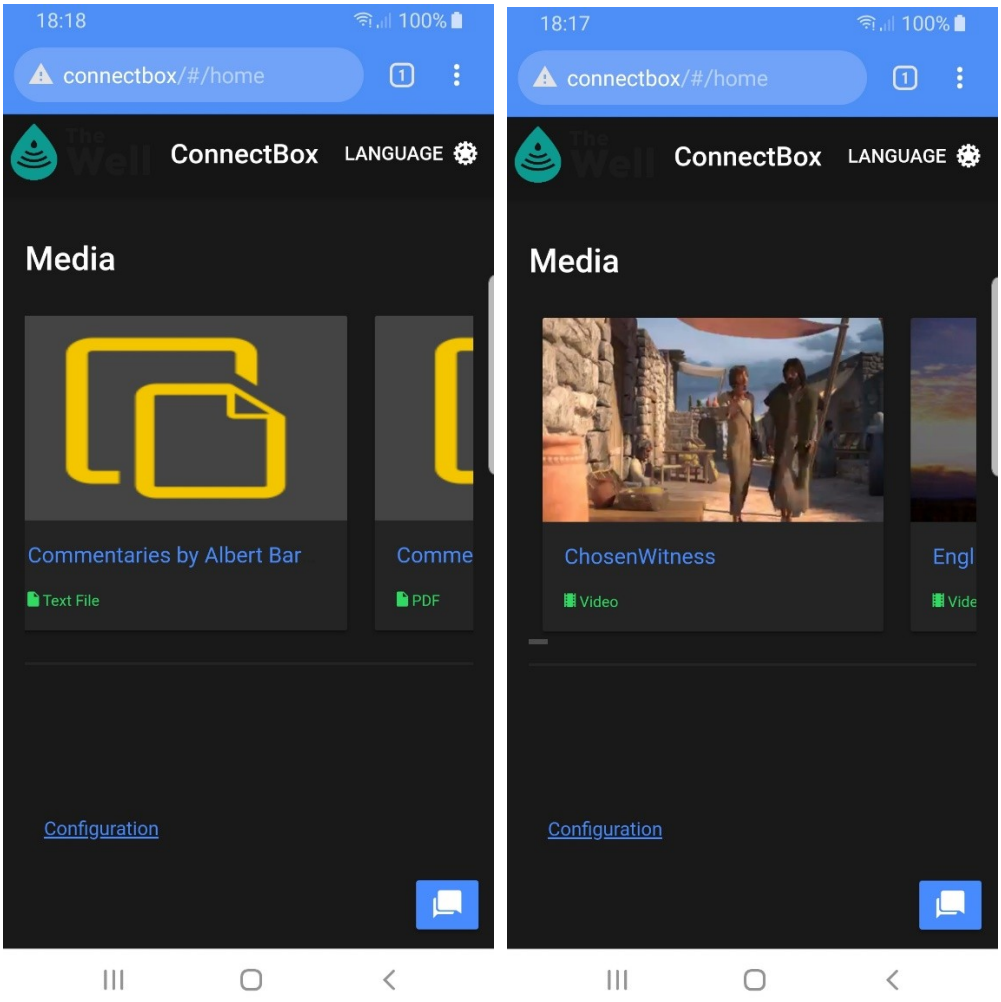


Note that the icons will also be replaced where possible by images of the media such as photo images, mp3 images, video frames at about 5seconds, etc.

For Languages there is a LANGUAGE dropdown to adjust the language you want to see, this will be selected automatically based on the user's browser language if it matches one of the Connectbox languages.



The interface will be generated on the first boot of the content being changed.



Note that to enable the ehanced interface thumbnails of images need to be generated to make the user experience more enjoyable. To do this when you place a new USB memory key in the unit it will mount it and then begin a process of collecting the media contents into thumbnails and screen format. While doing this you will notice on Firmware later than March 2023, a screen icon of the rotating hour glass.



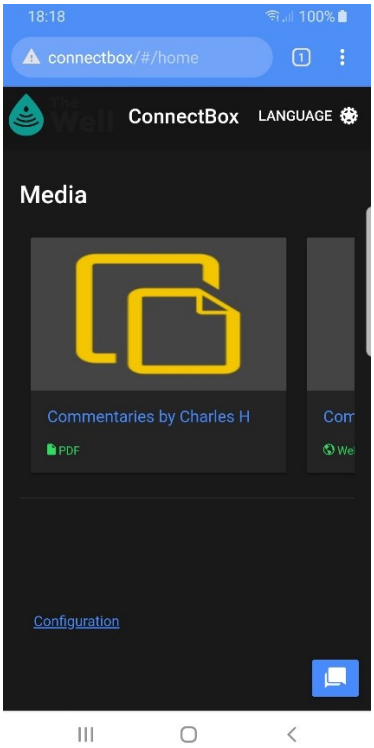
This will continue until the entire contents of the USB memory stick have been processed. This may take some time due to creatiion of lower resolution images but will only happen once and will create a file under the

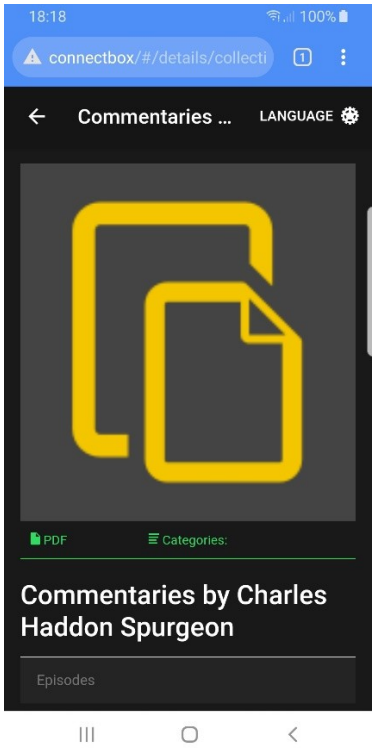
Root Direcctory/ content / saved.zip

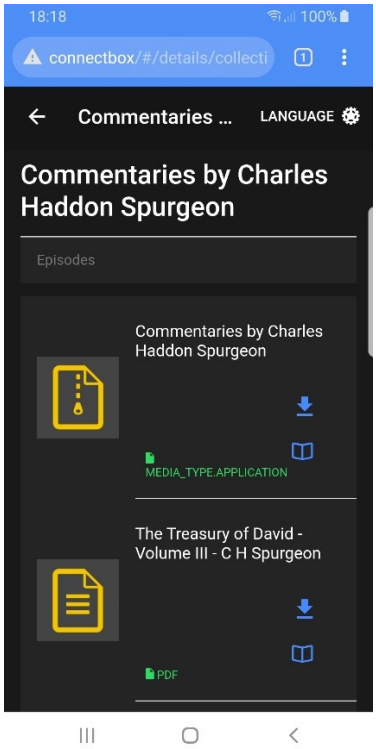
This saved.zip file contiains the quick load version of the interface. It will be subsequently used to load the interface when the USB is plugged in or the unit turned on to rapidly load the interface. If you add more content to the USB memory key then you would also need to delete this one file to ensure that the new content would be included in the user menu's.
It can take minits or up to several hours to do this the first time if you have large collections of media. The Connectbox creates zip files for media collections so that a download of the entire collection can be done. If the collection name.zip file exsists it will use this by default in the folder of the collection.

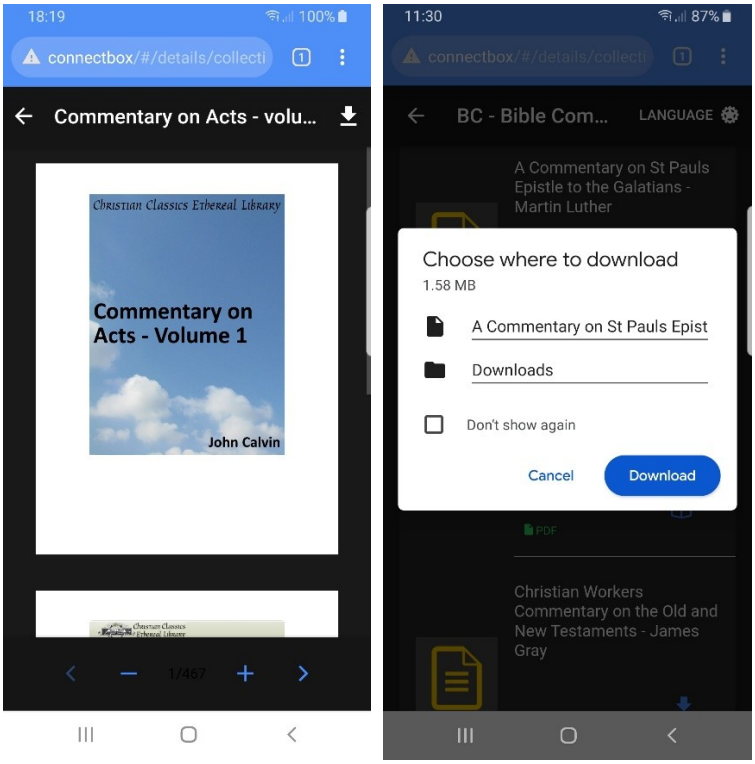
For elements of a collection:

Selection of the folder then provides a view of the content scroling up, showing the different types of content and either a view button or a download button. Selecting the view button will either show the media and allow you to interact with it or, you can select the download button to store it on your current device to look at later. Note that in this exaple there is a zip file of the Commentaries by Charles Haddon Spurgeon as the first item in the folder. Where the second item is a pdf file that can be opened with the built in pdf reader or downloaded.









To move backwards from your current location either cancel or use the upper left back arrow (sometimes you will find an upper right X to press to stop the media player then use the upper left back arrow. Additionally, you can always use the device back arrow button on the bottom or swipe back in some cases. In the case the file contains HTML media and an index.html file exists it will open in your browser that media directly. Navigation in this case is dependent on the application and you can always go back to starting screen by typing in the browser address bar http://connectbox or whatever the name of your connected system is. If all else fails type http://gowifi.org in the address bar.

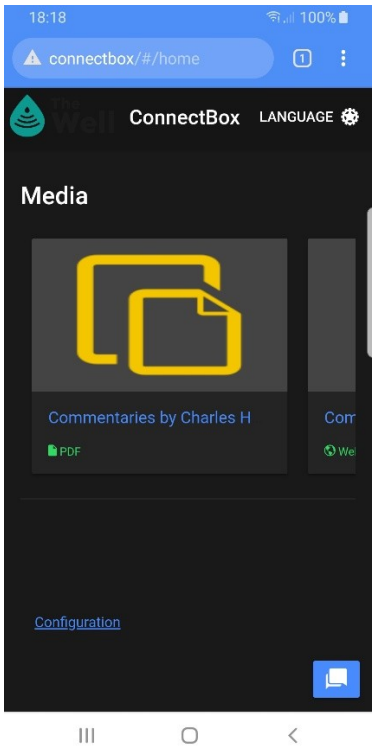
NOTE: That the Connectbox has a built-in pdf reader that renders pages in HTML format as well as a built in epub reader which also renders pages in HTML. This eliminates the need for mobile devices to have built in pdf or epub readers as applications. Note however that if you download the pdf, epub or zip file then the phone must have a built-in pdf, epub reader or unzip utility to use these files offline.

Step 4: Connecting your device to a running Connectbox

Once the connectbox has media and is sending WiFi signal (Green LED and Yellow LED on) you can connect your device to the connectbox Wifi signal. If you search on your device for local WiFi signals you should find the name of the Connectbox device. By default, this is "Connectbox" but this can be changed by the administrator of the box. So, know what WiFi signal name you are connecting to for the Connectbox. If the Connectbox has a barcode sticker on the top, then you can use your devices camera to read the information and make the connection. Note: on powerup it may take several minutes for the Green LED and subsequent Access Point name to start transmitting.

Once you have connected to the WiFi signal with your device, you may be asked if you want to use this device as it does not have internet access. You want to ensure you stay connected to the Connectbox WiFi signal even though it does not present internet access. This process may be different on different device. You will also generally be presented with a screen that tells you to open a web browser and connect to http://gowifi.org

After connecting open a web browser and enter in the URL bar **http://gowifi.org** this will then determine if your device is trying to use cellular internet to satisfy web requests or if it will communicate with the Connectbox. It may tell you to turn off your data connection. If so, please open the configuration window for WiFi and disable cellular data. Once completed it should take you directly to the Connectbox main page. In some cases, after turning cellular data off when asked you may need to refresh the page to get connected to the Connectbox main page.



Please know you cannot use **https**://gowifi.org as the Connectbox does not have security certificates that the browser will accept. So please use http://gowifi.org if your device tries to covert, it to https://gowifi.org it may give you a security warning. This is ok and you can proceed.

Note that the Connectbox main menu contains a language dropdown for selecting the language, a **configuration** link for the administration of the device (password protected), and possibly a chat link on the bottom right (if the administrator has not disabled this function) that allow you to chat with other users attached to the unit.

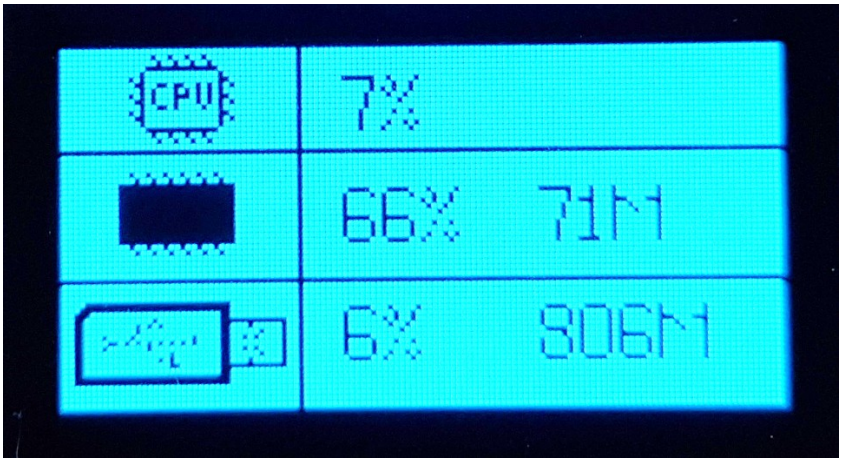
To administrate the Connectbox please see the Getting Started Guide for the appropriate version of software/hardware.

On Raspberry Pi units a monitor or TV connected to the HDMI port can show a playlist of the unit's videos, audio and pictures. It is set up in the HDMI Screen section of the Configuration page; see the Getting Started Guide.

Optional Step 5: Copying media from an external USB flash drive to the internal uSD flash drive.

When you are happy with the media content and have tested it using an external USB A flash drive plugged into the USB A port, you can copy this media in whole to the internal uSD memory of the unit, assuming there is enough free space.

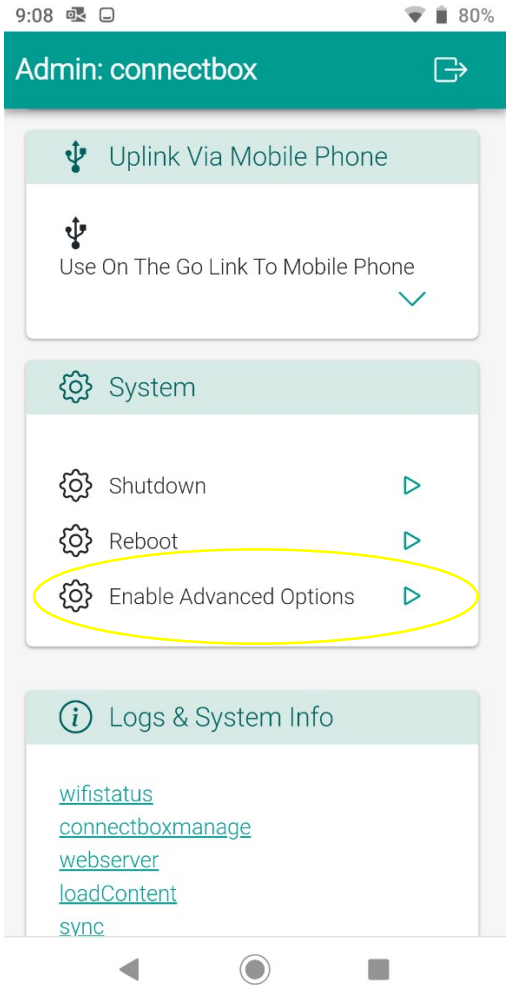
You can see the internal free space by using the front panel up arrow and navigating to screen 4 which shows the internal and external free space:



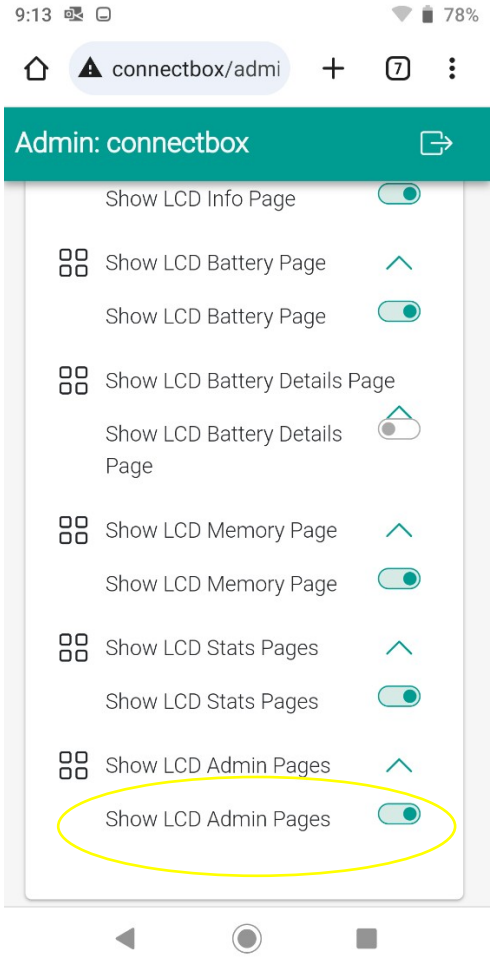
In this case the internal memory is 66% full using 71MB of data. The media storage is 6% used with 806MB of data. The media storage shows the internal uSD flash usage if no USB flash drive is plugged into the USB A port, and it show the amount of data on the USB-A flash drive if it is plugged in to the USB-A port. As mentioned above there is about 54 GB of available media storage on the internal uSD flash drive.

NEW: Copying a USB key to the internal uSD, or the internal uSD to a USB key, can also be done from the **Configuration** page once the advanced options are enabled: open **Box Content**, choose **Copy USB key to the box** or **Copy box content to a USB key**, press Start and confirm. The page shows each step (put the USB key in, Copying 45%, take the key out, Done) and the front panel shows the same screens. Erasing is only on the front panel. A large copy takes a while (about 20 minutes for 30 GB from a key, longer onto one); some very small keys get hot and drop out during a long copy, so use a full-size key and start the copy again after a Failed Copy.

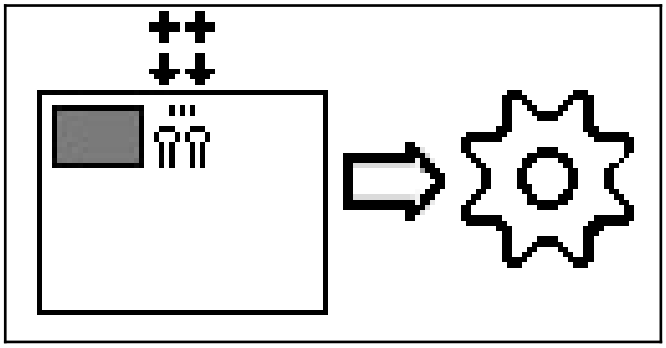
To erase the internal media storage on the uSD flash drive or copy from an external flash drive plugged into the USB-A port to the internal uSD flash drive you must have gone to the **configuration** setup and enabled **Extended Menus** then lower down on the configuration page **enable the advanced options.**



Otherwise, these menus are not available to the general user. Please see the Getting started guide for your device to be able to enable this feature. Then you need to scroll down on the screen to get to the screen menu to enable the LCD Admin Pages:



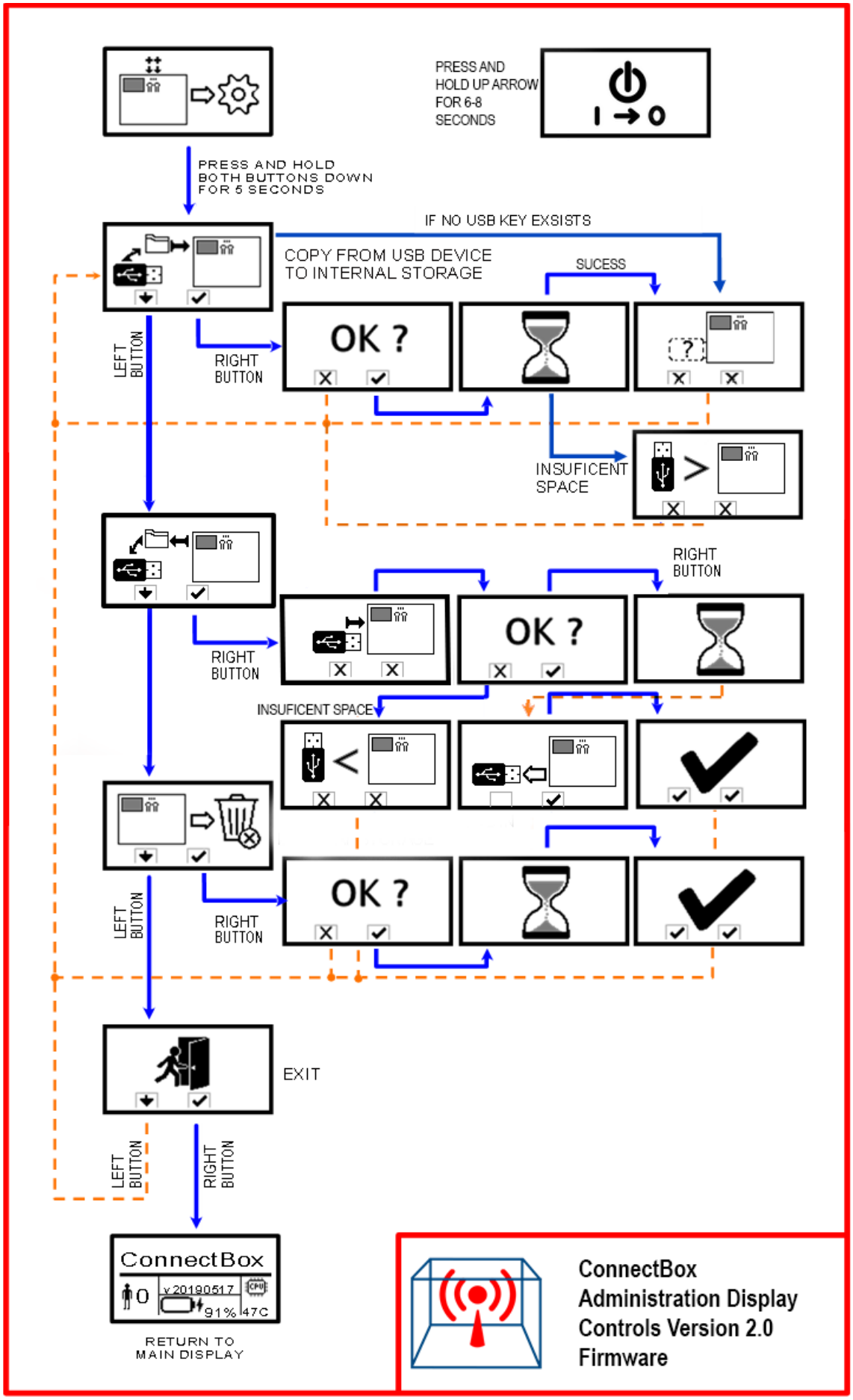
If the administrator menus are enabled then by pressing the down button to get to the 12th screen on the front panel which, will bring up the administrator entry screen showing how to enter file copy mode by pressing both up and down buttons simultaneously for more than 10 seconds.



If no buttons are pressed after 10 seconds, then the screen will go black again to conserve battery power. Pressing the Up button for more than 1 second will again show the first screen.

If at any active screen you push and hold both left and right buttons for more than 5 seconds you will open a file copy menu which is explained in more detail in the following block diagram:

NOTE: Every screen on the front panel can be enabled or disabled from the Configuration page and the extended menus enable page. The Administrative functions menu above is disabled by default and needs to be enabled in the Configuration page to be visible.
NEW: An IP addresses screen follows the statistics screens. It shows the unit's addresses: AP (the Wi-Fi phones join), WiFi (its link to another Wi-Fi network) and Eth (network cable).
The following diagram shows the functions of the extended administration menus on the front panel:


PRESS AND
HOLD UP ARROW
FOR 6-8
SECONDS
PRESS AND HOLD
BOTH BUTTONS DOWN
FOR 5 SECONDS
IF NO USB KEY EXSISTS
COPY FROM USB DEVICE
TO INTERNAL STORAGE
SUCESS
OK ?
LEFT
BUTTON
RIGHT
BUTTON
INSUFICENT
SPACE
RIGHT
BUTTON
OK ?
RIGHT
BUTTON
INSUFICENT SPACE
OK ?
LEFT
BUTTON
RIGHT
BUTTON
EXIT
LEFT
BUTTON
RIGHT
BUTTON
ConnectBox
v.20190517
91%
47C
RETURN TO
MAIN DISPLAY
ConnectBox
Administration Display
Controls Version 2.0
Firmware

Once you release the two buttons you will see the first page of the administrative screen. This is a copy function from the USB key to the internal uSD memory. If you click the right check mark button (the down arrow) it will take you to the, OK? Screen double checking you want to copy from your USB key to the internal memory. If you press the checkmark button (the down arrow) it will erase all previous content on the uSD then begin to copy all contents of your USB key to the internal memory if there is enough space (your uSD has about 54GB of capacity). If there is not enough space it will show the USB > connectbox image. Otherwise, you will get an hourglass symbol as it copies the media. Be prepared to wait some time as the copy is done at USB 2.0 speeds. If you have GB of data, it may take a long time. It's best to have the unit plugged in during this operation to ensure the battery doesn't discharge to far during the copy. Once the copy is complete it will display a Checkmark screen with double check's for the buttons. Or if you're not looking at the unit the display may be black (off). In either case the copy function is finished. Unplug the USB key and the connectbox will check for the content on the unit ensuring that all the contents are indexed.

The second screen of the admin functions is found by pressing the left button (the down arrow) on the copy screen reaching the Copy from Connectbox to USB memory key. This can be either a USB A memeory key or a uSD card in a USB A adapter plugged into the USB A port on the side. If you press the right button (down arrow) reaching the, OK? Screen if there is a memory card installed with enough space, otherwise it will display the insert memory card display until this is completed. This, OK? Screen double checks that you want to copy all the data files from the Connectbox to an external memory key. Note that if you press the right button to begin it will first erase all the content on the external USB A memory key screen, otherwise it will display the USB memory key < Connectbox capacity screen. If this happens you will need to get a larger storage key for the content. Note that without a USB A memory key installed you can see the size of the data storage from the front panel screens. Once the copy is started you will see the hourglass screen showing the copy operation in process. When successful it will display the check mark screen with double checkmarks for the buttons. Again, this copy process can take a long time so be patient.

The third screen of the admin functions is found by pressing the left button on the copy screen reaching the delete screen. This delete function is enabled by pressing the right button, the check mark. Once pressed it will ask for confirmation with an, OK? Screen. If you press the checkmark button (right down arrow) it will; proceed to delete all files on the data storage that are media. The hourglass will indicate the operation of the deletion of data files. When complete the checkmark screen will appear. **This process allows you to erase all data on the internal uSD. It is important to note that it will erase content on the storage device being used, either the internal uSD or the external USB A memory key.** Then by pressing either the right or left button you will return to the first copy screen.

If you are on the delete screen but do not want to delete the data on the storage space, pressing the continue button (left button) will take you to the Exit admin mode screen. If you press the continue button (left button) you will cycle back to the copy function. If you press the checkmark button (right button) you will exit the admin screens and be placed on the main page of the display

.

Once you have exited the enhanced menu's the administrator should turn off the enhanced screens using the configuration function on the main web interface.